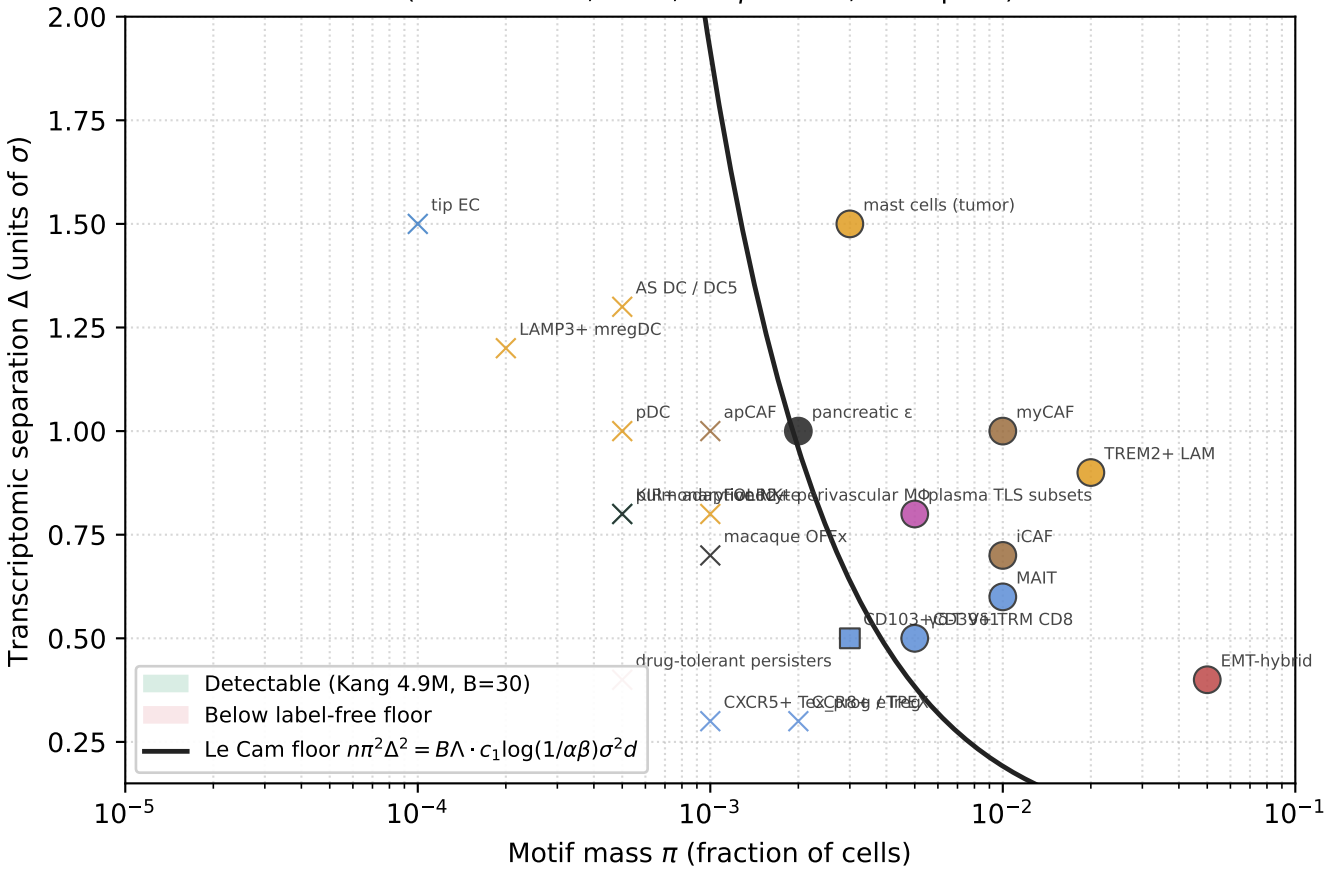

REAL detectability envelope — Kang 2024 pan-cancer atlas scale
($n = 4.9 \times 10^6$, $\Lambda = 3$, $\alpha = \beta = 0.05$, isotropic σ)
Transcriptomic separation Δ (units of σ)
Motif mass π (fraction of cells)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
tip EC
mast cells (tumor)
AS DC / DC5
LAMP3+ mregDC
pDC
apCAF
pancreatic ε
myCAF
TREM2+ LAM
perivascular Mφ
plasma
TLS subsets
iCAF
macaque OFFx
MAIT
CD103+CD39+ TRM CD8
drug-tolerant persisters
EMT-hybrid
CXCR5+ Tc
Detectable (Kang 4.9M, B=30)
Below label-free floor
Le Cam floor $n\pi^2\Delta^2 = B\Lambda \cdot c_1 \log(1/\alpha\beta)\sigma^2 d$